Temperature Monitor for Electronic Equipment



PETRE TZV. PETROV

As most electronics components' characteristics vary with temperature, they are selected as per expected operating temperature range of the equipment. So it is important that the equipment stays within the temperature range for which it is designed.

This circuit warns when the temperature reaches predefined danger levels. Here the predefined levels are set to 45°C, 65°C, 85°C and 105°C, but one can set any other temperature levels to suit the equipment in use.

TABLE I
Status of Presets and LEDs

Temperature	Adjusting preset	Status of LED
25°C	None	None
45°C	VR1	LED1
65°C	VR2	LED2
75°C	VR3	LED3
105°C	VR4	LED4

TABLE II
Test Points

Test point	Details
TP0	0V, GND
TP1	+5V
TP2	2.5V
TP3	Voltage corresponding to temperature

PARTS LIST

Semiconductors:
IC1 - TL431 shunt regulator
IC2 - LM324 quad operational amplifier
LED1-LED5 - 5mm LED

Resistors (all ¼-watt, ±5% carbon):
R1, R5, R18 - 1-kilo-ohm
R2, R7, R9-R12 - 10-kilo-ohm
R3, R20 - 4.7-kilo-ohm
R4, R19 - 2-kilo-ohm
R6, R17 - 560-ohm
R8 - 300-ohm
R13-R16 - 3-mega-ohm
R21-R24 - 680-ohm
VR1, VR2 - 200-ohm preset
VR3, VR4 - 100-ohm preset

Capacitors:
C1 - 0.033µF ceramic disk
C2 - 0.33µF ceramic disk
C3 - 10µF, 16V electrolytic
C4 - 100µF, 16V electrolytic

Miscellaneous:
NTC1 - 10k thermistor
CON1 - 5V DC connector
S1-S6 - On/off switch
- 14-pin IC socket
- 5V adaptor

Circuit and working

Fig. 1 shows the temperature monitoring circuit. It is built around 10k NTC thermistor NTC1, shunt regulator TL431 (IC1), popular comparator LM324 (IC2) and some other components.

IC2 activates corresponding LEDs (LED1 through LED4) when the temperature of thermistor NTC1 reaches the predefined level (refer Table I). The temperature is sensed by NTC1 and the produced corresponding voltage at point 'A' is fed to inverting terminal of all the four op-amps (A1 through A4). Switch S1 should be closed for this operation. This voltage level is compared with the reference voltage at non-inverting terminals of each op-amp. The reference voltage is obtained by voltage dividers from 2.5V reference source, which is produced using shunt regulator IC1. The reference voltage for each comparator can be set using presets VR1 through VR4.

The switches S2 through S6, together with resistors R2 through R6, are used for calibration purpose. For example, op-amp A4 of IC2 compares the voltage levels at point A (changing with change in temperature) with the reference voltage at its pin 12. The voltage level at pin 12 of A4 is adjusted

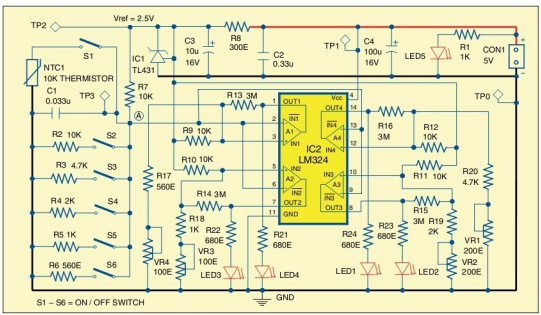


Fig. 1: Circuit of temperature monitor for electronic equipment